

Auteur: Seda Jusjajeva
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$$\begin{aligned}
 Df\left(\left(\frac{\pi}{3}, -\frac{2\pi}{3}\right), (4, -3)\right) &= \lim_{\lambda \rightarrow 0} \frac{f\left(\frac{\pi}{3} + 4\lambda, -\frac{2\pi}{3} - 3\lambda\right) - f\left(\frac{\pi}{3}, -\frac{2\pi}{3}\right)}{\lambda} \\
 &= \lim_{\lambda \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4\lambda\right) \cdot \cos\left(\frac{-2\pi}{3} - 3\lambda\right) - \frac{\sqrt{3}}{4}}{\lambda} \\
 &\stackrel{H}{=} \lim_{\lambda \rightarrow 0} \left(4 \cos\left(\frac{\pi}{3} + 4\lambda\right) \cos\left(\frac{-2\pi}{3} - 3\lambda\right) + \sin\left(\frac{\pi}{3} + 4\lambda\right) \left(-3 \sin\left(\frac{-2\pi}{3} - 3\lambda\right)\right)\right) \\
 &= 4 \cos \frac{\pi}{3} \cos \frac{-2\pi}{3} - \sin \frac{\pi}{3} 3 \sin \frac{-2\pi}{3} \\
 &= -1 - \frac{9}{4} \\
 &= -\frac{13}{4}
 \end{aligned}$$

References